

Curriculum

**B.Tech. In
Computer Science & Engineering**



**School of Computer Science and Technology
Faculty of Engineering and Technology
GM University
P. B. Road, Davanagere
Karnataka-577006**

Chancellor's Message

Welcome to GM UNIVERSITY with the above mission and an intense two and half decades of learning has inspired us to establish GM University that surpasses conventional learning, while revolutionizing new age education with modern practices and creating a holistic learning experience for each student at our campus, I wish each student at our campus along with their bright future a contributor to National Building



G.M. Lingaraju
Chancellor

I am immensely delighted in welcoming you to GM UNIVERISTY established through G.M UNIVERSITY ACT, 2022 received the assent of the Governor of Karnataka, a university of Unitary in nature Under Srishyla Educational Trust® Davanagere with rich history of its inception and legacy of

23 years in academic excellence and research mainly providing Quality education to Central Karnataka aspiring students. Under Srishyla Educational Trust last 2 Decades we have established 6 institutions to provide holistic in education in technical, management, Pharmacy, Pure science and commerce by applying best practices in teaching, learning with state of art infrastructure facilities.

6th July 2023 Government Order: ED/236/URC/2023 and on 21st August 2023, GM UNIVERISTY has become a reality, with Progressive inclusion all our institutions will be become part this University in near future.

GM UNIVERISTY started its journey with our founder's vision, a former Member of Parliament, Honourable Sri G Mallikarjunappa a visionary who believed in upgrading the livelihoods of people of this region, his efforts and reforms to improve the socio-economic conditions of Central Karnataka Educational Institutions are started, GM Group is committed to the growth of the region and has invested in education, and industries that can contribute to the growth of the district. In our Lifetime for future endeavours becoming a University is a necessary part of the journey,

We at GM University believe in nurturing the talent of young minds by providing best of

Curriculum, innovative teaching methodology, state-of-art infrastructure, new age technology, result oriented international collaborations & tie ups, Job Oriented Industry partnerships, Expertise panel, Real-time Research & innovations, Incubations to contribute to nation and Start-ups to Build Entrepreneurs, we build all this on the inspiration of our founder Honourable Sr. G. Mallikarjunappa. Diversity and inclusivity are two of our major strengths to create an eco-system where in all Stake holders are beneficiaries a Student, a Parent, an Institution at large our Nation.

In the year 2000 we started with small number of students after 23 Year of experience and academic excellence with GM University inception we are looking forward to make 20,000 Students Campus in near future with the need to prepare our students for rapidly changing & interconnected world we constantly strike to bring into campus a global perspective as the students are aiming at global careers. As a university we aim to create a socially responsible global citizen who can efficiently lead organisations, society and country. GMU with its Eco-system and leadership getting equipped to contribute better to New Age education, industry and academia.

I thank all students, parents, faculties, researchers and advisors for your support and encouragement with I invite you once again to be part of future GM UNIVERSITY Journey.

**G.M. Lingaraju Chancellor,
GM University**

Vice Chancellor's Message

Universities stand as vibrant crucibles, where the alchemy of knowledge creation and dissemination shapes the very fabric of civilized society. It's not just about feeding minds; it's about nurturing holistic development—imbuing students with emotional intelligence, social acumen, physical vitality, economic savvy, environmental consciousness, and spiritual resonance. Think of it as a recipe for crafting global citizens, ready to make



Dr. S. R. Shankapal
Vice Chancellor

waves in the scientific, technological, cultural, social, healthcare, and economic realms of our interconnected world. In this grand symphony, the university curriculum, guided by our stellar faculty, takes centre stage, shaping students into responsible global citizens.

Welcome to GM University, a rising star aspiring for global acclaim! We're not just offering programs; we're crafting experiences. Each program is a bespoke journey, meticulously tailored to meet the aspirations of our dynamic students and their families. Our faculty, a league of well-educated maestros with real-world expertise, employs a symphony of pedagogical approaches—traditional, project-based, and problem-based learning. We bridge the gap between theory and practice, creating an immersive learning experience. Learning at GM University is not just about books; it's an exhilarating adventure, supported by cutting-edge infrastructure designed for well-rounded student development.

India, with its robust economy, plays a pivotal role globally, and our graduates are set to contribute significantly to this narrative. GM University, founded by the visionary Shrishaila Educational Trust under the dynamic leadership of Sri. G. Mallikarjunappa and Chairman G M Prasanna Kumar, steered by the visionary Chancellor Sri. G M Lingaraju, aspires not just to be a global educational hub but to redefine higher education in the central part of Karnataka. It's more than an institution; it's a commitment to transforming lives in the region.

As Ron Lewis, the American politician, aptly puts it, 'Ensuring quality higher education is one of the most important things we can do for future generations.' GM University, with its unwavering commitment to excellence, invites you to embark on a transformative learning journey within our vibrant campus. We extend a warm welcome and wish you the very best

Dr. S. R. Shankapal
Vice-Chancellor, GM University

Pro Vice Chancellor's Message

GM University started its journey as per the Karnataka State University Act 19(2023) and is still in a formative stage, but it has inherited a long history from GM Institute of Technology founded in the Year 2000. A warm and affectionate welcome to all the students who aspire to be part of the intellectual fraternity of GM University, Davanagere.



Dr. H.D. Maheshappa
Pro Vice Chancellor

This is the start of your academic journey- one that is bound to give you quality learning for your bright future. The University has a mission for different academic bodies. It is working tirelessly for academic expansion and diversification, widening the academic canvass in horizontal and vertical dimensions. By now the knowledge- standards on the campus has been meeting the intellectual requirements of the society, in the areas of Engineering, IT, Sciences and, Commerce Management.

The University puts in sustained efforts in developing a contemporary and relevant curriculum, meeting and beating various benchmarks of quality assurance, developing a healthy academic governance structure, cutting edge curriculum, management blended with updated research and support.

The University marches ahead to attain its mission through a goal of blending contemporary knowledge with skills and values with an intention to groom high calibre thinkers, morally upright and physically sound, citizenry, inspired with a proactive approach to take up challenges confronting the present society and develop productive individuals in the fast- changing knowledge society. But the core objective here at GM University is to work for the upliftment of humanity at large.

We have full faith in your capabilities. During your stay in the University, your talent will be honed and you will be exposed to a cutting edge, competitive environment. We wish you rise up to become successful leaders in your chosen areas by your zealous participation in the corporate life. We wish you good luck.

Dr. H.D. Maheshappa
Pro Vice Chancellor, GM University

Registrar's Message

The GM University has a history of being more than 22 years, the great human being and philanthropist Sri. G Mallikarjunappa is a founder of the institution with the vision to provide a quality education to the rural community. In pursuance to this it is our unwavering commitment to the highest quality education that has sped us along on this journey. I would also like to thank the generations of parents who have entrusted us with the care and education of their children.



Dr. Sunil Kumar B.S.
Registrar

It is their faith and belief in the GMU values – Perseverance, Excellence and Service – that has helped us come this far. I believe that education has the most critical role to play in the progress of a nation. We realize that our students represent the future of our society and we take our responsibility seriously.

We ensure that the rock-solid foundation we help them build here – both, in terms of skills and values – will stand them in good stead no matter which career they choose. Along with the requisite academic and related skills, we try and instil empathy for the less fortunate in our students and a concern for the environment around them.

We create opportunities not just for young people to learn, but also for them to experience the fulfilment that comes from sharing this learning with others. At GMU, we will always be focused on delivering education that will help create the leaders of tomorrow's world – men and women who will not just be successful in their individual capacities, but who will in turn, contribute to and nurture the world they live in.

**Dr. Sunil Kumar B.S. Registrar,
GM University**

Senior Dean's Message

I extend a warm welcome to all the students who have recently become part of the esteemed GM University in Davangere. GM University boasts a tranquil and conducive environment, providing students with the ideal setting to pursue their studies and excel academically. The university actively encourages energetic and creative students to engage in research and innovation, addressing societal challenges and fostering entrepreneurial ventures. The unwavering passion of our students to be

inventive and actively participate in industry and society is well-supported by our highly qualified and experienced faculty members.

In addition to ensuring academic activities are carried out with the utmost efficiency, GM University provides guidance and mentorship to students, enabling them to publish research papers, file patents, and achieve new standards of excellence. Our Central Library houses a wealth of printed and e-books, research journals, and reference materials of global standards.

As an academically autonomous institution, GM University takes the liberty to incorporate the latest research outcomes from related areas into the curriculum; ensuring students are well-prepared for the industry. This approach makes education at GM University more meaningful, pragmatic, and diverse. It also opens avenues for greater public and industry access to the university's scientific and research achievements, promoting the practical applications of these outcomes in public utility and industrial productivity.

GM University is committed to supporting interdisciplinary research, facilitating the identification and development of new fields of knowledge crucial for the bright future of our graduates, postgraduates, and Ph.D. degree holders. Our humble contributions aim to create an intellectual base for the sustainable economic development of the nation.

We believe that the scholars graduating from GM University will be valuable assets to global society. We wish them all a bright future and a peaceful life ahead.

Dr. Karibasappa Kwadiki



Senior Dean - Faculty of Engineering and Technology, GM University

Dean's Message

I am delighted to have this opportunity to address you and share some insights about our institution. It is my privilege to serve as the Dean, and I am proud to lead a diverse and dynamic academic community.

Our university is committed to providing a world-class education that fosters critical thinking, innovation, and personal growth. As you browse through our website, you will find information about our exceptional faculty members, the wide range of academic programs we offer, and the vibrant campus life that awaits our students.

Our faculty members are experts in their respective fields, dedicated to imparting knowledge, and passionate about nurturing the intellectual curiosity of our students. They are committed to academic excellence and are constantly engaged in cutting-edge research and scholarly activities. By joining our university, you will have the opportunity to learn from some of the brightest minds in their disciplines.

We take great pride in our student-centered approach, which acknowledges the unique aspirations, talents, and needs of each individual. We believe that education should go beyond the classroom, and thus, our students are encouraged to actively participate in extracurricular activities, student organizations, and community engagement initiatives. These experiences enhance personal growth, cultivate leadership skills, and foster a sense of social responsibility.

At our university, we embrace diversity and inclusivity, recognizing that it is the foundation for a rich and vibrant academic environment. We value and respect the diverse perspectives and lived experiences of all individuals. It is through the exchange of ideas and collaboration across different cultures and backgrounds that we can truly thrive as a community.

As you navigate our website, you will find information about our state-of-the-art facilities and library resources. Our commitment to providing a conducive learning environment ensures that our students have access to the tools and resources they need to excel in their studies. Whether you are a prospective student, a parent, an employer seeking talent, or a curious visitor, I invite you to explore our website, learn more about what sets us apart, and envision yourself as part of our community. Should you have any queries or need further information, please do not hesitate



to reach out to us. We are here to assist you every step of the way.

Dr. S V Prakash

Dean - Faculty of Engineering and Technology, GM University

Director's Message

Welcome to the School of Computer Science and Technology – a hub of innovation, exploration, and academic excellence. As the Director of this dynamic School, it is my pleasure to extend a warm welcome to students, faculty, and all enthusiasts passionate about the limitless possibilities that the world of computer science offers.



**Dr. Sanjay Pande M
B**

At our school, we strive to create an environment that nurtures creativity, critical thinking, and technological innovation. Our distinguished faculty, comprised of experts in various domains of computer science, is dedicated to provide a transformative learning experience, through a blend of rigorous academics, hands-on projects, and industry collaboration. We prepare our students to be leaders in the ever-evolving field of technology.

As technology continues to reshape the world, our school remains at the forefront of innovation and development. We encourage our students to be active contributors to this transformative journey, pushing the boundaries of what is possible.

Your journey at the School of Computer Science and Technology will be characterized by exposure to cutting-edge technologies, collaboration with peers and industry professionals, and the opportunity to turn your ideas into reality.

I invite you to explore our website, learn about our programs, faculty, and research initiatives. Whether you are a prospective student, a current member of our community, or a visitor curious about the world of computer science, I am confident that you will find the School of Computer Science and Technology to be a place where innovation thrives and futures are shaped.

I look forward to witnessing your success and contributions to the exciting world of computer science.

Warm regards,

Dr. Sanjay Pande M B

Director, School of Computer Science and Technology

HOD's Message

Dear Students of the Department of Computer Science and Engineering,
As we embark on another exciting academic year, it is my great pleasure to extend a heartfelt welcome to each of you within our vibrant community of Computer Science and Engineering.



Our department stands as a bastion of innovation, where cutting-edge research and education converge to shape the future of technology. Together, we are poised to explore new horizons, tackle emerging challenges, and pioneer advancements that will redefine the landscape of computing.

Whether you are a seasoned faculty member, a dedicated staff member, or an eager student, your contributions are instrumental in driving our department's mission forward. As we strive for excellence in teaching, research, and service, let us cultivate an environment that nurtures creativity, collaboration, and inclusivity.

In the pursuit of knowledge, let us remain steadfast in our commitment to pushing the boundaries of what is possible. Let us embrace diversity of thought, embrace the spirit of inquiry, and embrace the transformative power of technology to positively impact society.

As we embark on this journey together, I am confident that our collective efforts will lead to extraordinary achievements and lasting contributions to the field of Computer Science and Engineering.

I am excited to witness the incredible accomplishments that lie ahead and to celebrate our continued success as a dynamic and forward-thinking department.

Warm regards,

Dr. B N Veerappa

Head -Computer Science and Engineering

G M University

Acknowledgement

The successful development of our program is a testament to the collaborative efforts and unwavering support of a diverse group of individuals who have played pivotal roles throughout the journey. We extend our heartfelt gratitude to the following:

Program Development Team: A profound appreciation goes to the dedicated and dynamic members of the program development team. Your collective expertise, creativity, and tireless commitment have been instrumental in conceptualizing and shaping a program that stands as a beacon of academic excellence.

Faculty Members of the Department: Our sincere thanks to the esteemed faculty members of the department for your invaluable contributions. Your academic insights, subject matter expertise, and enthusiasm have played a crucial role in crafting a curriculum that meets the highest standards of quality and relevance.

Faculty Members of the School: We express our gratitude to all faculty members of the school for your unwavering support and engagement. Your diverse perspectives and commitment to fostering a vibrant academic community have been instrumental in shaping the program into a comprehensive and enriching educational experience.

Members of Board of Studies: A special acknowledgment to the members of the Board of Studies for your guidance and strategic input. Your collective wisdom and experience have been pivotal in ensuring that the program aligns with the latest developments in the field and meets the evolving needs of our students.

Members of Academic Council: Our sincere thanks to the members of the Academic Council for your oversight and endorsement. Your commitment to academic rigor and excellence has provided a solid foundation for the program's integrity and credibility.

Director of the School: We extend our gratitude to **Dr. Sanjay Pande M B** for your visionary leadership and steadfast support. Your guidance has been instrumental in navigating the complexities of program development, and your commitment to fostering a culture of innovation

is truly commendable.

Senior Dean and Dean of the School: Our sincere thanks to **Dr. Karibasappa Kwadiki** and **Dr. S V Prakash** for your leadership and unwavering commitment to academic excellence. Your support has been crucial in ensuring the alignment of the program with the broader goals and mission of the school.

Registrar: A special acknowledgment to **Dr. Sunil Kumar B S** for your meticulous coordination and administrative support. Your attention to detail has been essential in facilitating the smooth development and implementation of the program.

Pro-Vice Chancellor: Our gratitude extends to **Dr. H.D. Maheshappa** for your guidance and support at the university level. Your commitment to advancing education and promoting innovation has played a crucial role in the success of our program.

We would like to express our sincere gratitude to **Sri Y. U. Subhash Chandra**, Management Representative, for his unwavering support while realising GM University.

We extend our deepest gratitude and heartfelt appreciation to **Dr. K. Divyananda** and **M. Rafiq Raj** of GM Group for their unwavering dedication, tireless efforts, and visionary leadership during the establishment of GM University.

Vice-Chancellor and Chancellor of the University: Our deepest appreciation to **Dr. S.R. Shankapal** and **Sri. G M Lingaraju** for your visionary leadership and steadfast support. Your commitment to the pursuit of knowledge and academic excellence has provided the foundation for the development of this program.

Together, your collaborative efforts have not only shaped an outstanding program but have also contributed to the advancement of education within our institution. We are truly grateful for your invaluable contributions and unwavering support.

With sincere thanks,
Dr. SHIVANAGOWDA G M
Head - Computer Science & Engineering Department

Preface

Dear Program Participants,

It is with great pleasure and enthusiasm that I present to you the curriculum document for our esteemed program. As the Head of the Program, it is both an honour and a responsibility to guide you through a transformative learning journey.

This curriculum has been meticulously designed to not only impart knowledge but also to cultivate critical thinking, creativity, and problem-solving skills. Our commitment to excellence in education is reflected in the comprehensive structure of this program, which aims to empower you with the tools and insights necessary for success in a rapidly evolving world.

In developing this curriculum, we have drawn upon the latest advancements in the field, ensuring that you receive cutting-edge knowledge and skills that are relevant to the current and future landscape. Our dedicated team of members of faculty and industry experts has worked tirelessly to create a curriculum that not only meets academic standards but also exceeds expectations, preparing you for the challenges and opportunities that lie ahead.

Throughout the program, you will engage in a dynamic blend of theoretical learning and practical application, fostering a holistic understanding of the subject matter. The inclusion of real-world case studies, interactive projects, and collaborative activities will provide you with hands-on experience, enhancing your ability to apply theoretical concepts in practical scenarios. As you embark on this educational journey, I encourage you to approach each course with curiosity and a thirst for knowledge. Our program is not just about acquiring information; it is about developing a mind-set of continuous learning and adaptability. The skills you gain here will serve as a foundation for a lifetime of intellectual growth and professional achievement.

I extend my sincere gratitude to the faculty, staff, and everyone involved in the development of this curriculum. Their dedication and expertise have been instrumental in shaping a program that we believe will inspire and empower you to reach new heights.

I am confident that your time in this program will be rewarding and transformative. Embrace the challenges, seize the opportunities, and let this curriculum be the catalyst for your personal and professional success.

Wishing you an enriching and fulfilling learning experience.

Warm regards,

Dr. Shivanagowda G M

Head - Computer Science & Engineering Department

Members – Board of Studies

S. No.	Name	Designation and Affiliation
1.	Dr. H.D. Maheshappa	Pro-Vice Chancellor, GM University
2.	Dr. Karibasappa Kwadiki	Senior Dean Faculty of Engineering and Technology, GM University
3.	Dr. S V Prakash	Dean Faculty of Engineering and Technology, GM University
4.	Dr. Sanjay Pande M B	Director School of Computer Science and Technology, GM University
5.	Dr. Shreedhara K S	Professor, University BDT College of Engineering
6.	Dr. Manjunatha S	Professor, Department of Computer Science and Engineering, Cambridge Institute of Technology, Bangalore - 560036
7.	Dr. Rajendra Hegadi	Professor and Dean, Dept. of Data Science and intelligent Systems, IIIT Dharwad
8.	Dr. C M Ravi Kumar	Professor, University BDT College of Engineering, Davangere - 577004
9.	Dr. Veerappa B N	Professor Dept. of CSE, GM University, Davangere – 577006
10.	Dr. Shivanagowda G M	Professor Dept. of CSE, GM University, Davangere – 577006
11.	Dr. Asha K	Associate Professor Dept. of AIML, GM University, Davangere – 577006
12.	Dr. Neelambike S	Associate Professor Dept. of ISE, GM University, Davangere – 577006
13.	Mrs. Nayana K	Assistant Professor Dept. of CSE Engineering, GM University, Davangere – 577006
14.	Mrs. Spoorthi B	Assistant Professor Dept. of Civil , GM University, Davangere – 577006
15.	Mr. Santoshkumar M	Assistant Professor Dept. of CSE Engineering, GM University, Davangere – 577006
16.	Mrs. Sandhya R S	Assistant Professor Dept. of CSE,,GM University,,Davangere – 577006

17.	Mrs. Kavayashree P N	Assistant Professor Dept. of CSE, GM University, Davangere – 577006
18.	Dr. Varun B K	Associate Professor Dept. of Civil, GM University, Davangere – 577006
18.	Mrs. Muktha Pujjar	Assistant Professor Dept. of AIML, GM University, Davangere – 577006

Members – Academic Council

S. No.	Name	Designation and Affiliation
1.	Dr. S R Shankapal	Vice Chancellor, GM University
2.	Dr. H.D. Maheshappa	Pro-Vice Chancellor, GM University
3.	Dr. Sunil Kumar B S	Registrar, GM University
4.	Dr. Karibasappa Kwadiki	Senior Dean Faculty of Engineering and Technology, GM University
5.	Dr. S V Prakash	Dean Faculty of Engineering and Technology, GM University
6.	Dr. Basavaraj Swamy	Dean, GM University
7.	Mrs Swetha Marigoudar	Dean, GM University
8.	Dr. Santhosh B M	Dean, GM University
9.	Dr. Sanjay Pande M B	Director, GM University
10.	H. A Balasubramanya	Director (Formerly), RMP
11.	Dr. Deepak K T	IIT Dharwad
12.	Dr. Sreedhara K S	University BDT College of Engineering, Davangere
13.	Mr. virupaksha	Alumnus of GMIT
14.	Sunil S Batnavar	ALS Bangalore
15.	Girish Pattanshetu	An Independent Journalist
16.	Dr. Shashibala Sigh	Director - NIPER
17.	Dr. Mahadeva Prasanna	IIT Dharwad
18.	Anil Kumar Ramaseshan	CTO, TE-Connect
19.	Deviprasad Karnik	Scientist
20.	Prof. Shankar Murthy	Dean-Academics, IIT Mumbai
21.	Prof Somashekar T S	NLSIU-Bangalore
22.	Dr. M Somashekar	Former Principal, RL Law College, Davangere